SOFR Cap Forward Volatility, Risk Premia, and Rates-Volatility Carry

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Abstract

This paper asks whether the forward-volatility curve implied by SOFR caps behaves like an expectations curve or like a compensated risk-premium curve. I strip caplet-level forward normal volatility from quoted SOFR cap flat vols, validate the curve construction against an independent benchmark workbook, and then test whether longer forward-vol points forecast future short forward volatility. The evidence is deliberately conservative: stripped forward vols display a large positive premium relative to later 0.5Y forward vols and forward realized SOFR volatility, but overlapping horizons and the short post-2022 sample make naive significance claims fragile. A walk-forward ridge forecast converts the premium into a stylized rates-volatility carry screen. The screen earns positive after-cost proxy P&L in the tested sample, with annualized Sharpe 2.41 after leakage-safe volatility targeting. Purged forecast validation, CPCV-style strategy folds, and block-permutation nulls reduce the risk that the result is a path-fit artifact, and the block-bootstrap median path remains positive, but the block-bootstrap lower interval still crosses zero. I therefore interpret the result as a research-grade investability test, not as a production caplet trading strategy. The production conclusion is narrower and stronger: SOFR cap forward vols contain an economically meaningful premium, and a disciplined researcher should only promote that premium to a trade after caplet-level marks, bid/ask, margin, and execution are modeled explicitly.

1 Introduction

SOFR caps quote a flat volatility for the full option package, but the trading object inside the package is a sequence of caplets. This distinction matters because the market can price near-term policy uncertainty, medium-term volatility risk, and long-run volatility supply through different parts of the curve. A cap flat vol is therefore a par object. A stripped forward vol is the marginal object. It asks how much volatility the next caplet must carry, after the earlier caplets have already been priced, in order for the full cap to match the market quote.

This paper studies that marginal object. The empirical question is not simply whether SOFR cap volatility is high or low. The question is whether the forward-vol curve is mainly an expectations curve for future short-tenor volatility, or whether it embeds a premium that can be monitored and possibly harvested. That framing is important for trading. If the curve is an unbiased expectations curve, then a high long-forward vol is mostly a forecast. If the curve contains a persistent premium, then the long-forward point may be rich relative to future realized volatility even when the directional level of rates is difficult to forecast.

The empirical contribution is deliberately disciplined. First, I build the caplet curve from the quoted cap surface rather than taking the forward-vol object as given. The implementation bootstraps a quarterly SOFR discount curve, converts normal cap vols into Black-equivalent flat vols, reprices each cap as a sum of caplets, and solves the residual marginal caplet volatility. Second, I test the expectations-hypothesis analog with horizon-matched HAC standard errors and effective non-overlapping sample counts. Third, I compare implied forward vol with forward realized SOFR volatility and policy-cycle regimes. Fourth, I build a walk-forward trading screen that uses only lagged information, transaction costs, and volatility targeting. Fifth, I stress that screen with purged and embargoed cross-validation, CPCV-style strategy folds, feature ablations, and block-permutation null tests.

The main finding is a premium rather than a clean forecast. The longer forward-vol points are far above later realized short forward vols, while the forecasting slopes are far from one. A stylized weekly carry rule that sells rich forward vol and buys cheap forward vol earns 1291.21 bp of proxy P&L over 124 backtest observations after a 2 bp turnover cost and lagged risk scaling. The result is strongest in the pause and easing regimes. It is not enough evidence to claim executable alpha, because the test is short, the product-level mark-to-market is simplified, and the block-bootstrap P&L interval crosses zero at the lower end. The correct reading is that the premium is economically interesting and worth expanding, not that the current notebook is a deployable options book.

The remainder of the paper is organized as follows. Section 2 places the design in the rates-vol and empirical-inference literatures. Section 3 describes the data and caplet stripping procedure. Section 4 defines the forecast, premium, and strategy objects. Section 5 reports the main results. Section 6 reports robustness and anti-overfit checks. Section 7 states the implementation gap. Section 8 concludes, and Appendix A gives a claim-by-claim audit of what the repo proves.

2 Related Literature

The curve-construction layer follows the standard logic of caplet stripping in interest-rate option markets. A cap can be valued as a portfolio of caplets under the Black framework [1], and the forward-volatility curve is obtained by solving the marginal caplet vols that reproduce quoted flat cap prices. The conceptual analogy is the same as stripping zero rates from par swap rates: the market quote is a bundled price, while the research object is the sequence of marginal forward prices.

The empirical layer is closer to tests of expectations hypotheses and risk premia than to option pricing alone. A forward rate, forward volatility, or forward variance can be read as an expectation plus compensation for bearing risk. The point of the paper is to separate those two readings as far as the sample permits. Newey–West inference is used because overlapping forecast horizons create serially correlated forecast errors [3]. The strategy section is intentionally closer to a trading research workflow: it treats the premium as a candidate signal, then asks whether the signal survives out-of-sample forecasting, costs, and position scaling.

3 Data and Curve Construction

3.1 Data

The local data set contains 989 SOFR cap-volatility quote dates from 2022-03-17 through 2025-12-31, SOFR swap quotes, and daily SOFR reference rates. The raw Excel workbooks are not tracked in Git. The public repository contains the executed notebook, reusable Python functions, tests, documentation, figures, and this paper.

The core workbooks are `project_cap_vol_ts.xlsx`, `cap_curves_2025-06-30.xlsx`, and `ref_rates.xlsx`. The first file supplies the time series of cap flat vols. The second supplies the validation date used to verify the curve implementation. The third supplies SOFR reference rates used for realized-volatility benchmarks. Optional FRED data can be downloaded by the repo script for policy-regime context, but those CSV files are written under `data/raw/` and remain outside version control.

3.2 From Flat Cap Volatility to Forward Caplet Volatility

Let T index cap maturity and let $i = 1, \ldots, n(T)$ index quarterly caplets. A market flat vol σ_T^{flat} is the single volatility that prices the full cap:

$$C(T, K, \sigma_T^{\text{flat}}) = \sum_{i=1}^{n(T)} Z_i B(F_i, K, \sigma_T^{\text{flat}}, t_i), \tag{1}$$

where Z_i is the discount factor, F_i is the forward SOFR rate, K is the cap strike, and $B(\cdot)$ denotes the Black caplet price. The stripped forward-volatility problem solves a different equation. After all previously stripped caplets have been valued, the next unknown caplet volatility σ_i^{fwd} must match the remaining market value:

$$C(T_i, K, \sigma_{T_i}^{\text{flat}}) - \sum_{j<i} Z_j B(F_j, K, \sigma_j^{\text{fwd}}, t_j) = Z_i B(F_i, K, \sigma_i^{\text{fwd}}, t_i). \tag{2}$$

Quoted normal vols are converted into Black-equivalent flat vols using the ATM approximation

$$\sigma_T^B \approx \frac{\sigma_T^N}{F_T}. \tag{3}$$

The stripped Black forward vols are then converted back into normal-volatility units for the time-series analysis. This conversion is not a theoretical claim that the Black and normal models are identical. It is an implementation bridge that allows the validation workbook and the time series to be read in consistent units.

The 2025-06-30 validation workbook is used as a hard implementation check. The maximum absolute errors are small relative to the curve levels:

Validation object	Max absolute error
Discount factors	0.000018
Forward rates	0.000125
Forward Black vols	0.002311

4 Empirical Framework

4.1 Expectations-Hypothesis Regression

The first empirical object is an expectations-hypothesis analog for forward volatility. For short tenor $\delta = 0.5$ and longer tenor τ, let $h = \tau - \delta$. The regression is

$$\sigma^N_{t+h,\delta} = \alpha_\tau + \beta_\tau \sigma^N_{t,\tau} + \varepsilon_{t+h,\tau}. \tag{4}$$

If the longer forward-vol point were an unbiased forecast of future short forward vol, the slope would be near one after accounting for measurement and risk-premium terms. In this sample, the forecast horizons overlap heavily. The paper therefore reports HAC standard errors with lags matched to the horizon and an effective non-overlapping sample count.

This section is intentionally not framed as a structural pricing test. The sample starts in 2022, which means the observations are concentrated in a hiking cycle, a pause, and the beginning of an easing regime. That is useful for studying the modern SOFR market, but it limits the independent time-series variation available for long-horizon tests.

4.2 Forward-Volatility Risk Premium

The second object is the ex post forward-volatility term premium,

$$\mathrm{VTP}_{t,\tau} = \sigma^N_{t,\tau} - \sigma^N_{t+h,0.5}. \tag{5}$$

This is a realized premium measure: it compares the long forward-vol point observed at time t with the short forward-vol point eventually observed at $t + h$. I also compare implied forward vol with forward realized SOFR volatility. That realized-vol benchmark is economically important because option sellers ultimately care about realized rate volatility, not only about later implied volatility marks.

4.3 Walk-Forward Trading Screen

The strategy layer asks whether the premium can be converted into a disciplined research signal. Let $\widehat{\sigma}^{RV}_{t+1}$ denote an expanding-window forecast of forward realized SOFR volatility using information available at t. The signal is

$$q_t = \sigma^N_{t,0.5} - \widehat{\sigma}^{RV}_{t+1}. \tag{6}$$

The strategy sells forward vol when q_t is above 8 bp, buys forward vol when q_t is below -8 bp, scales exposure to a volatility-risk target, and charges 2 bp of normal-vol transaction cost on turnover. The forecasting features are current implied vol, trailing realized SOFR volatility, forward-vol slope, 13-week vol momentum, and 13-week vol-of-vol. Ridge regression is used for stability because the feature set is collinear and the sample is short.

This design is deliberately less ambitious than a production trade. It is a screen for whether a lagged volatility-risk-premium signal has enough structure to justify deeper modeling. A production implementation would require caplet-level marks, executable bid/ask, margin, liquidation rules, and a definition of the traded cap/floor structure.

5 Results

5.1 The Stripped Surface

The stripped forward-normal-volatility panel is sampled weekly. The front end is the most volatile part of the surface because it is closest to monetary-policy uncertainty, while the longer end is smoother and behaves more like a premium state variable.

Tenor	Mean bp	Std bp	Min bp	Max bp
0.5Y	72.88	38.07	15.44	255.21
1.0Y	109.05	37.58	55.73	242.63
1.5Y	134.54	36.46	68.45	221.48
2.0Y	136.15	32.84	73.42	198.39
3.0Y	128.37	30.69	74.47	195.88
5.0Y	103.90	18.24	69.88	178.16

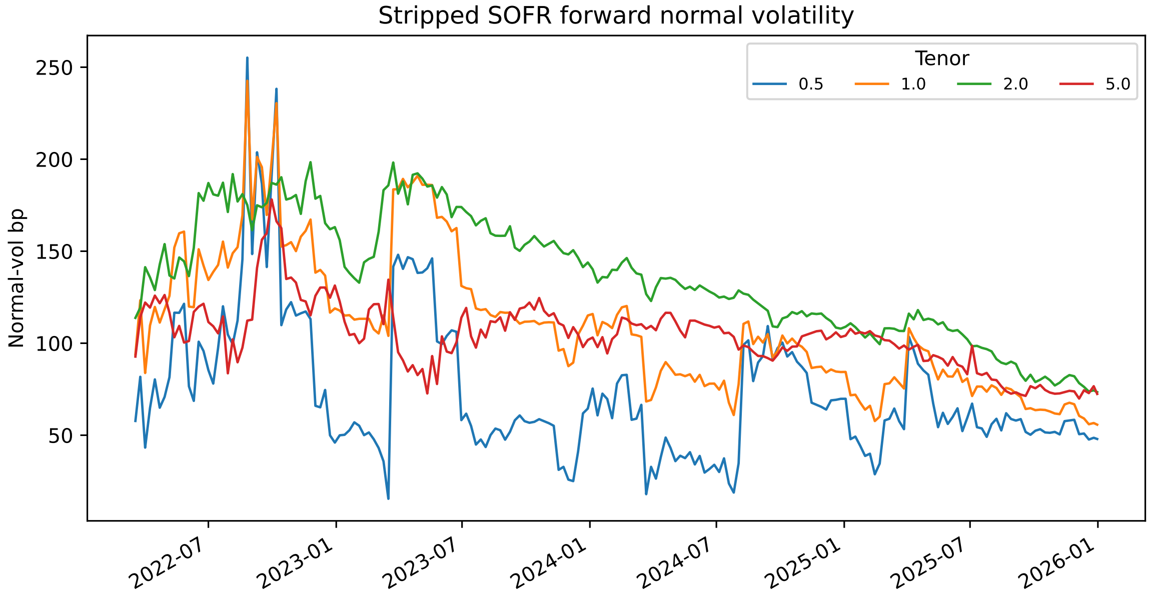

Figure 1: Stripped SOFR forward normal volatility by tenor. The front end reacts most sharply to policy uncertainty, while the longer end is smoother and more naturally interpreted as a premium state variable.

A PCA of weekly forward-vol changes is included in the notebook as a surface diagnostic rather than as a trading model. Its purpose is to show whether the surface mostly moves in parallel or whether slope and hump movements matter. This turns the project from a one-curve bootstrap into a rates-volatility-surface study.

5.2 Expectations-Hypothesis Evidence

The expectations-hypothesis slopes are far from one:

τ	Alpha bp	Beta	HAC s.e.	R^2	Effective N
1.0Y	54.82	0.117	0.140	0.015	6
1.5Y	103.02	-0.260	0.198	0.074	2
2.0Y	22.93	0.225	0.096	0.057	1
3.0Y	35.06	0.177	0.077	0.026	1

The result should be read with two constraints in mind. First, the slopes are economically inconsistent with an unbiased expectations curve in this sample. Second, the effective independent sample shrinks quickly as the horizon extends. The paper can therefore claim strong descriptive evidence of a forward-vol premium, but it cannot claim a precise long-horizon structural estimate.

5.3 Term Premium and Realized-Volatility Benchmark

The forward-volatility premium is positive across the main tenors:

τ	Mean bp	Std bp	Positive fraction	Effective N
1.0Y	47.11	46.98	90.75%	7
1.5Y	84.94	46.88	95.24%	3
2.0Y	99.15	25.47	100.00%	2
3.0Y	96.03	21.69	100.00%	1

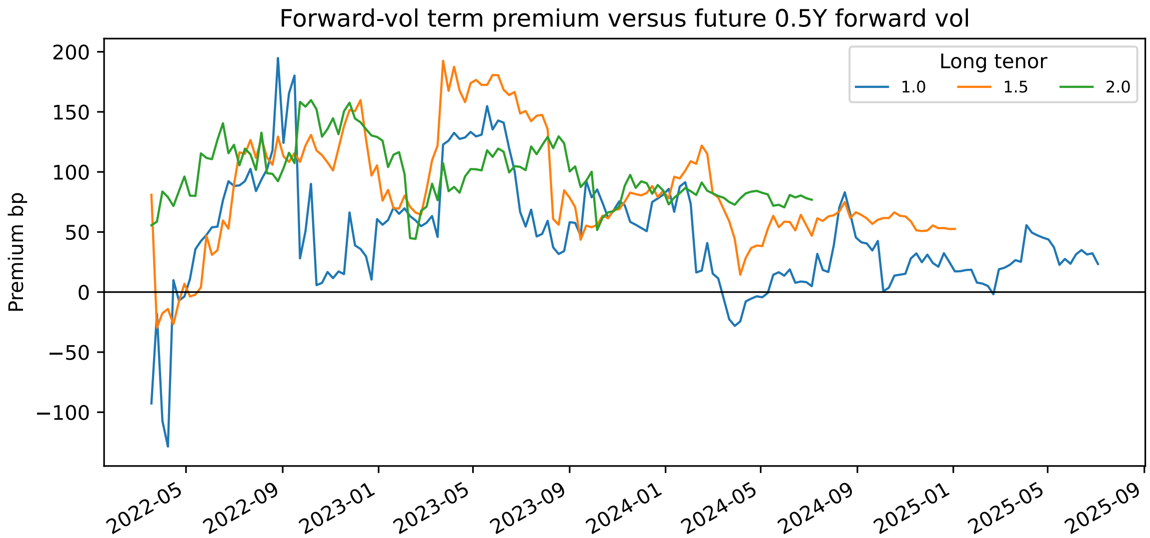

Figure 2: Forward-vol term premia relative to future 0.5Y forward vol. The premium is persistent, but overlapping horizons mean that weekly observations overstate the amount of independent evidence.

The realized-volatility benchmark tells the same story in a different language. Comparing 1Y forward normal vol with forward realized SOFR volatility over 3M and 6M windows gives a mean volatility risk premium of roughly 32 bp, with a positive fraction near 70%. This is economically plausible in a rates market where volatility sellers demand compensation for policy uncertainty, but it is still a premium diagnostic rather than an executable P&L statement.

5.4 Strategy Screen

The walk-forward strategy turns the premium into an investability screen. It trades only after the minimum training window, uses an expanding ridge forecast, scales exposure by risk, and subtracts turnover costs. The headline diagnostics are:

Diagnostic	Value
Backtest observations	124
Total stylized P&L	1291.21 bp
Annualized Sharpe	2.41
Max drawdown	-434.03 bp
Hit rate while active	70.30%
Turnover	15.00

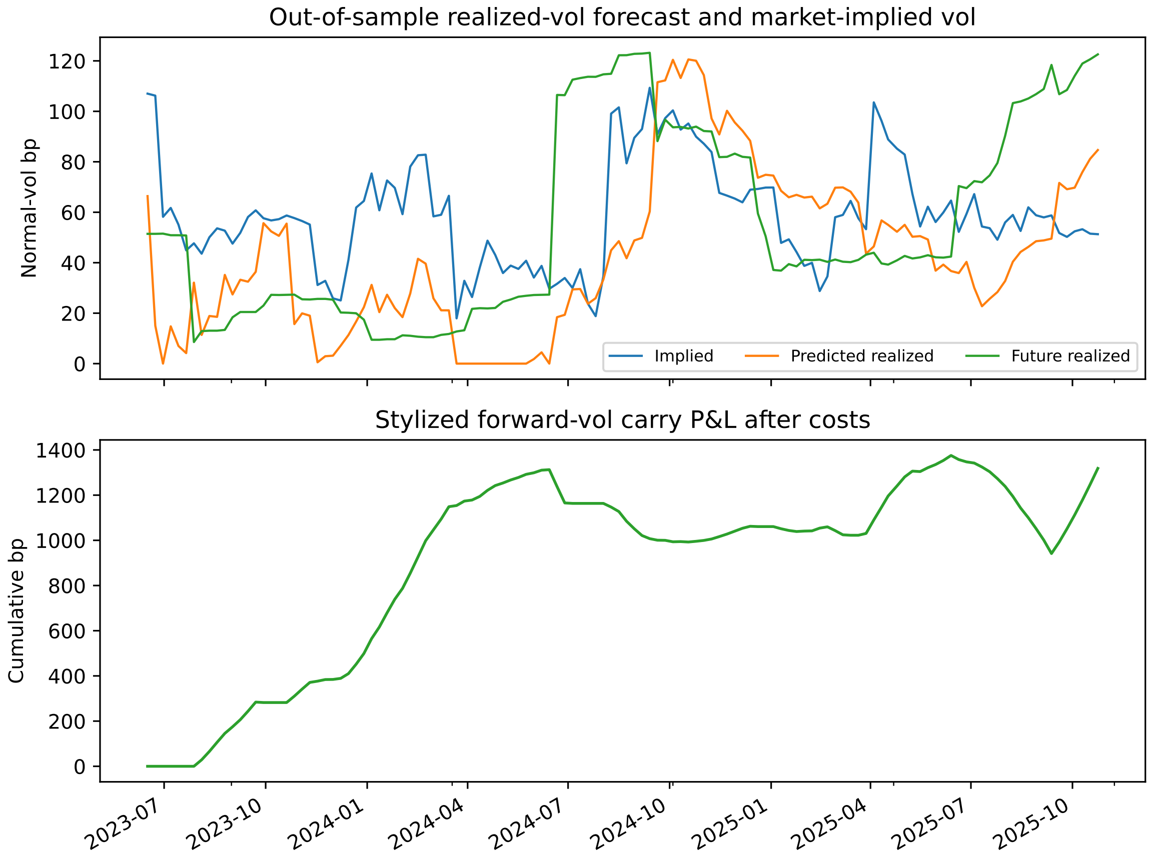

Figure 3: Out-of-sample realized-vol forecast, implied forward vol, and stylized carry P&L after turnover costs. The strategy is an investability screen for the premium, not a complete caplet execution model.

The result is promising, but its interpretation is intentionally bounded. P&L is concentrated in the pause and easing regimes; the hiking regime has no active observations after the minimum training window. The volatility-targeting scale is computed from lagged forecast errors, so the position size at a signal date does not use the current realized-vol label. A block bootstrap for total P&L gives a wide interval, from roughly -286.69 bp to 2605.65 bp, and the corresponding Sharpe interval runs from -0.58 to 5.41. The median bootstrap endpoint is 1160.82 bp, close to the mean bootstrap endpoint of 1149.68 bp, so the positive result is not just a mean-path artifact. That does not erase the economic signal. It prevents the paper from treating the signal as already production-ready.

6 Robustness and Anti-Overfit Controls

The strongest source of overfitting risk is the short, overlapping-label structure. A 63 business-day forward realized-vol label creates dependence across adjacent weekly observations. The notebook therefore avoids shuffled cross-validation and uses purged, embargoed chronological folds. For forecast validation, the sample is split into six chronological groups, two groups are held out at a time, the label horizon is purged by 13 weekly rows, and a two-week embargo is imposed around test blocks. The full feature set has the best mean purged-CV RMSE, 58.85 bp, versus 65.71 bp for implied-plus-realized features and 68.92 bp for implied vol alone. The result supports using the additional curve and volatility-state variables, but the improvement is modest enough to keep the claim conservative.

The strategy is tested across pre-declared configurations rather than optimized on the full path. Thresholds of 5, 8, and 12 bp, risk targets of 25 and 35 bp, transaction costs of 2 and 4 bp, and two ridge penalties are evaluated with CPCV-style folds. The best average CPCV Sharpe is about 1.91, and the top configuration has positive total P&L in 80% of folds. The base configuration remains near the top of the table rather than appearing as a single lucky parameter choice.

Finally, a block-permutation null breaks the alignment between the signal and future realized-vol labels while preserving local label dependence. This is a nonparametric null: it does not assume Gaussian forecast errors or thin-tailed volatility shocks. The observed total P&L of 1291.21 bp has a one-sided block-permutation p-value of 0.049; the observed Sharpe has p-value 0.037. The null mean and median total P&L are 91.59 bp and 76.39 bp, and the null 95th percentile is 1285.73 bp, just below the observed value. This is supportive but not overwhelming. The correct interpretation is that the signal survives the first serious anti-overfit battery, while the bootstrap uncertainty and short regime history still require a product-level execution study before any production alpha claim.

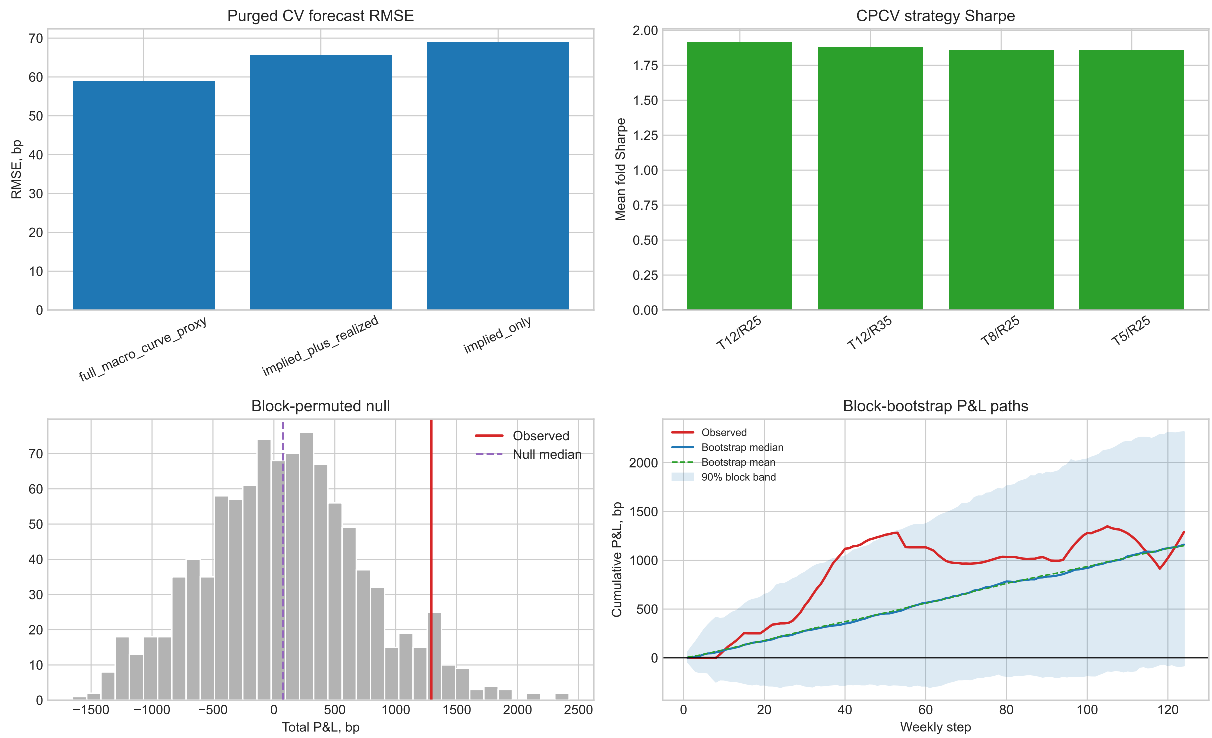

Figure 4: Robustness evidence for the SOFR rates-volatility carry screen. The panels show purged forecast-CV RMSE by feature set, CPCV strategy Sharpe for pre-declared configurations, and a block-permuted null distribution for total P&L, plus observed, mean, and median block-bootstrap cumulative P&L paths.

7 Implementation and Trading Interpretation

The clean research object is forward normal volatility. The tradable object would be a caplet, cap, floor, swaption overlay, or a structured package with actual bid/ask and margin. That mapping is not mechanical. A production version must mark each caplet through time, specify the strike and delta convention, incorporate skew if the trade is not exactly ATM, model realized hedge P&L, and charge realistic costs. It must also decide whether the economic exposure is short vol, long convexity, curve-vol relative value, or a carry strategy with drawdown constraints.

The current repo is therefore production-grade as a research artifact, not as a live trading system. It demonstrates a complete workflow: instrument stripping, validation, surface diagnostics, forecasting, cost-aware backtesting, regime attribution, and a clear stop before overclaiming. A quant researcher or trader should be able to inspect the notebook, reproduce the paper tables, and see exactly what still needs to be built before the idea becomes executable.

8 Conclusion

This paper builds a SOFR cap forward-volatility curve and uses it to study whether the market's marginal caplet vol is a forecast or a premium. The empirical evidence supports the premium interpretation. Forward vols are persistently above later short forward vols and forward realized SOFR volatility, while the expectations-hypothesis slopes are far from one.

The trading evidence is useful but deliberately not oversold. A lagged ridge forecast and a cost-aware carry screen generate positive stylized P&L, but the sample is short, the independent horizon count is small, and the product-level execution model is not yet complete. The robustness layer makes the project harder to dismiss as path fitting, but it does not remove the need for instrument-level execution. The correct next step is not to make the language stronger. It is to make the implementation stronger: caplet marks, bid/ask, margin, skew, executable structures, and stress tests across rate-vol regimes.

A Claim-by-Claim Methodology Audit

Claim	Implemented evidence	Allowed interpretation
Forward vols are stripped from cap quotes	Cap flat vols are repriced as caplet portfolios; validation errors are reported for discount factors, forwards, and Black vols.	The repo implements the caplet-stripping workflow rather than relying on a precomputed forward-vol surface.
The curve is not an unbiased expectations curve	Forecast slopes for future 0.5Y forward vol are far from one, with HAC standard errors and effective sample counts.	Strong descriptive evidence of a premium; not a large-sample structural test.
The premium is economically meaningful	Forward-vol premia are positive across tenors and realized-vol comparisons show roughly 32 bp mean premium.	The market compensates sellers of forward rates volatility in this sample.
The strategy is researchable	Walk-forward ridge forecast, lagged features, lagged risk scaling, costs, regime attribution, and bootstrap uncertainty are implemented.	The signal justifies deeper trading research; it is not yet a caplet execution strategy.
The result is not just a path-fit artifact	Purged forecast CV, CPCV-style strategy folds, feature ablations, sensitivity tables, and block-permutation null tests are implemented.	The evidence is robust enough for a research artifact, but the bootstrap interval still crosses zero.
The public repo is safe to share	Raw workbooks and downloaded public data live outside version control; tests and documentation state the data contract.	Recruiters can review methodology and code without private data or secrets being published.

B How to Read the Repository

Start with this paper for the research object and claim hierarchy. Then read `SOFR_Cap_Forward_Volatility.ipynb` for the executed analysis. The reusable implementation is in `src/sofr_forward_vol.py`; tests are in `tests/test_sofr_forward_vol.py`; the data contract is in `data/README.md`; and the local knowledge-base protocol is in `docs/knowledge_base_protocol.md`.

References

[1] Black, F. (1976). The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2):167–179.

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[3] Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.